ECON 1450

Consumer and producer surplus

Chapter 4

Today

- Exam overview/study strategies
- Chapter 4/Practice

Friday

- Remaining CH4 content
- Exam review (focus on CH3 & 4)

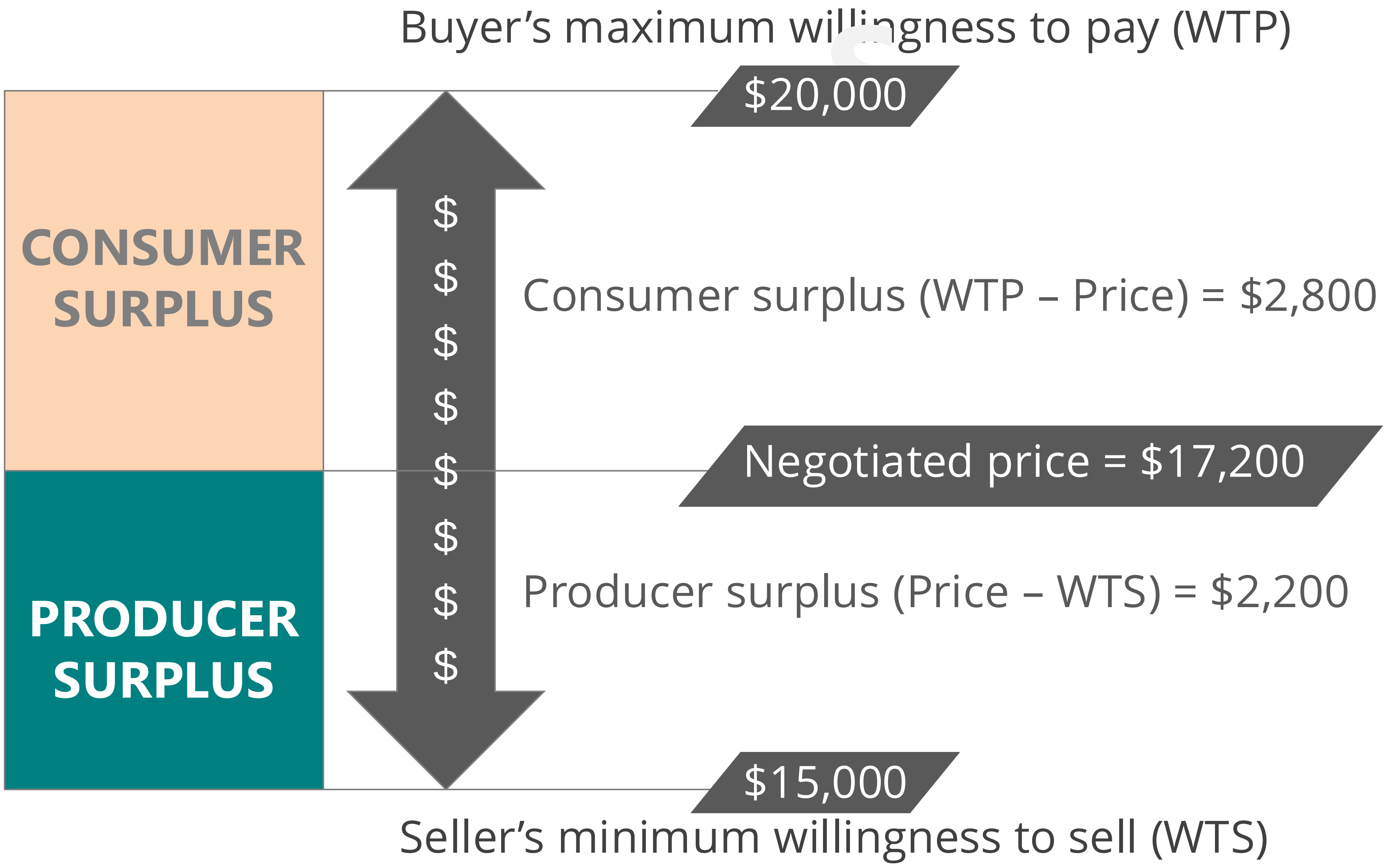
Where are we?

- We just covered **eleven principles of economics** **(CH1) semi - asynchronously**
- We learned how to set up **supply and demand curves** and **move them** around **(CH3).**

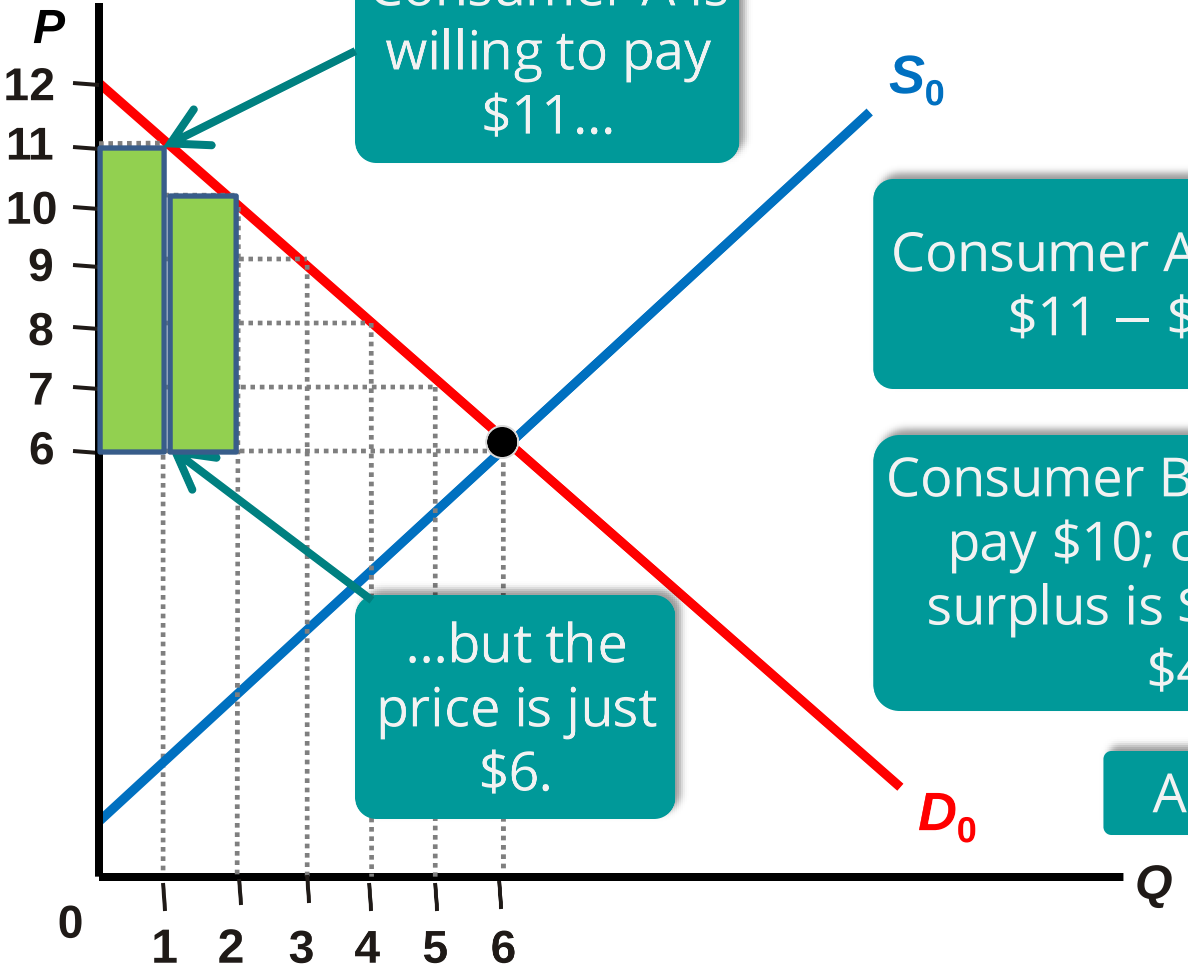
Learning objectives

- Calculate **consumer** and **producer surplus**
- Define and identify **deadweight loss**
- When markets lead to **efficiency**
- Types of **market failures**

CONSUMER & PRODUCER SURPLU



INDIVIDUAL CONSUMER



Consumer A is willing to pay $11…

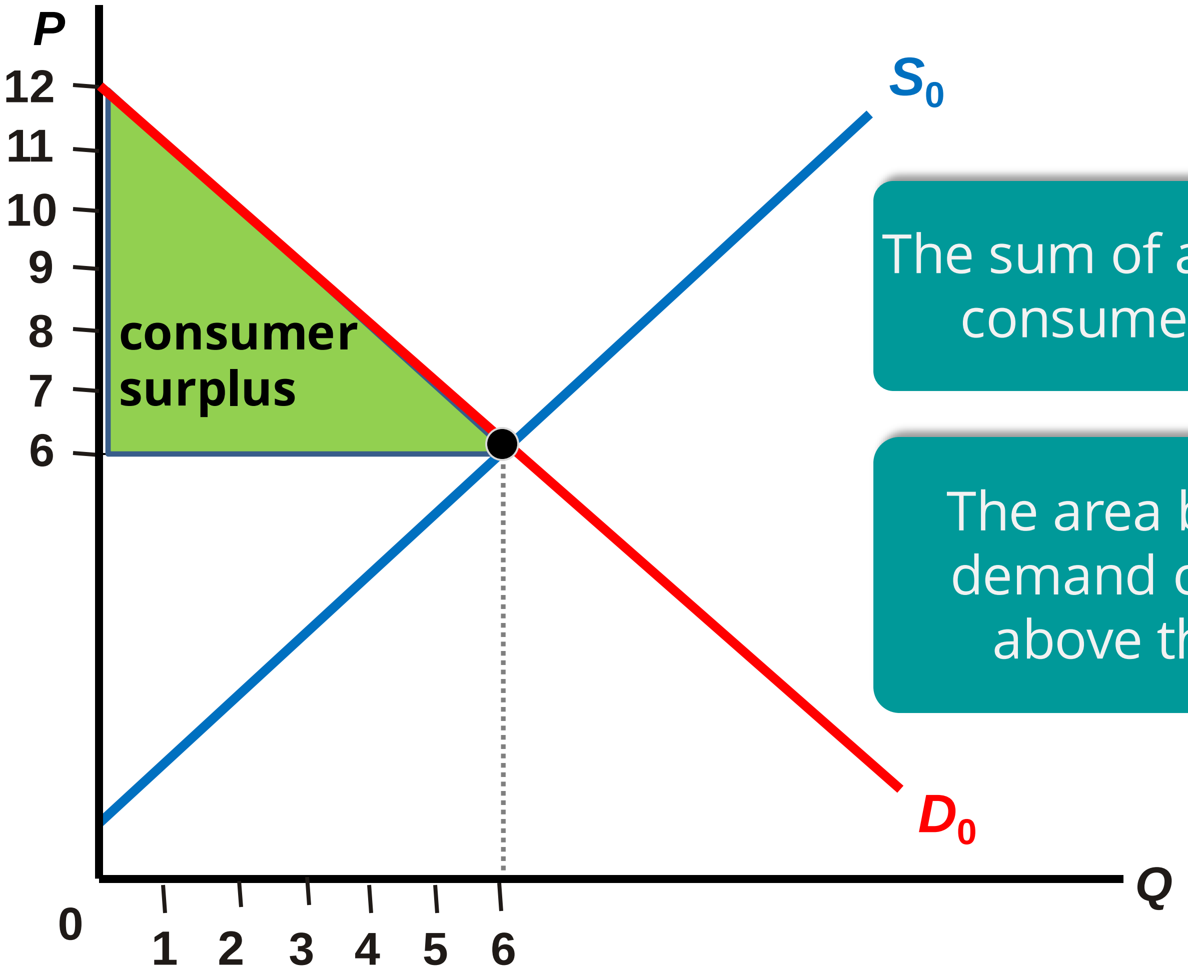
…but the price is just $6.

Consumer A's surplus is $11 – $6 = $5.

Consumer B is willing to pay $10; consumer surplus is $10 – $6 = $4.

And so on…

TOTAL CONSUMER SURPLUS



The sum of all individual consumer surplus

The area below the demand curve and above the price

Buy some new shirts, George!

George wants to buy some new shirts for work. He is willing to pay $35 for the first shirt, $25 for the second shirt, and $15 for the third.

Oxford & Co. (his favorite shirt shop!) sells shirts for $28 each.

What is the efficient number of shirts for George to buy?

a) 0
b) 1
c) 2
d) 3

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What is George's consumer surplus?

a) $0
b) $7
c) $10
d) $28
e) $35

Calculating consumer surplus

Mark and Rasheed are buying new calculators for Math 19. Mark is willing to pay $75 and Rasheed is willing to pay $100 for a graphing calculator. The price for a calculator at the bookstore is $65. How much is their total consumer surplus?

a) $10
b) $35
c) $45
d) $60

Figure: The Market for Hamburgers

If the price of hamburgers is $2, consumer surplus will equal:

a) $650
b) $400
c) $225
d) $450

If the price of hamburgers is $2.50 today, what will happen to the price in the coming days/months?

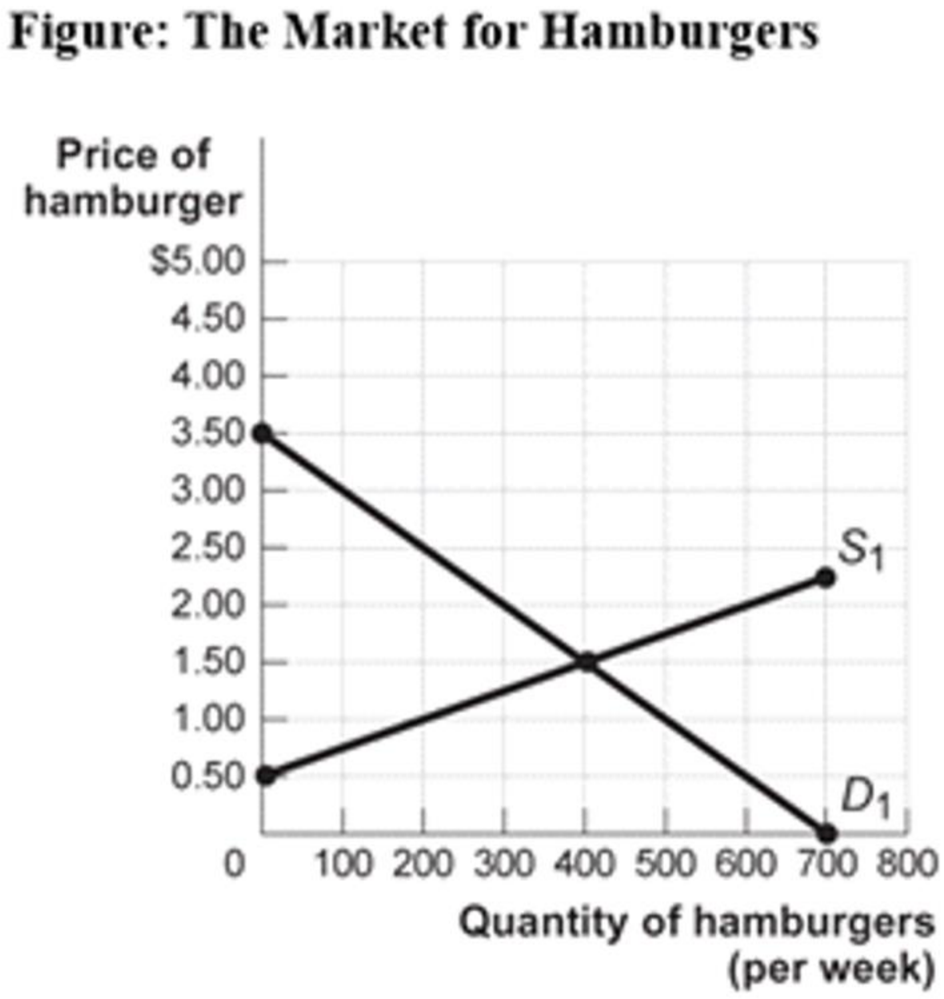
a) It will fall

b) It will rise

c) It will remain the same

d) Not enough information

If the price of hamburgers rises, what will be the main effect on the chart?



a) Supply will shift left
b) Supply will shift right
c) Demand will shift left
d) Demand will shift right

If the price of a hamburger is $2, consumer surplus will be …

a) $650

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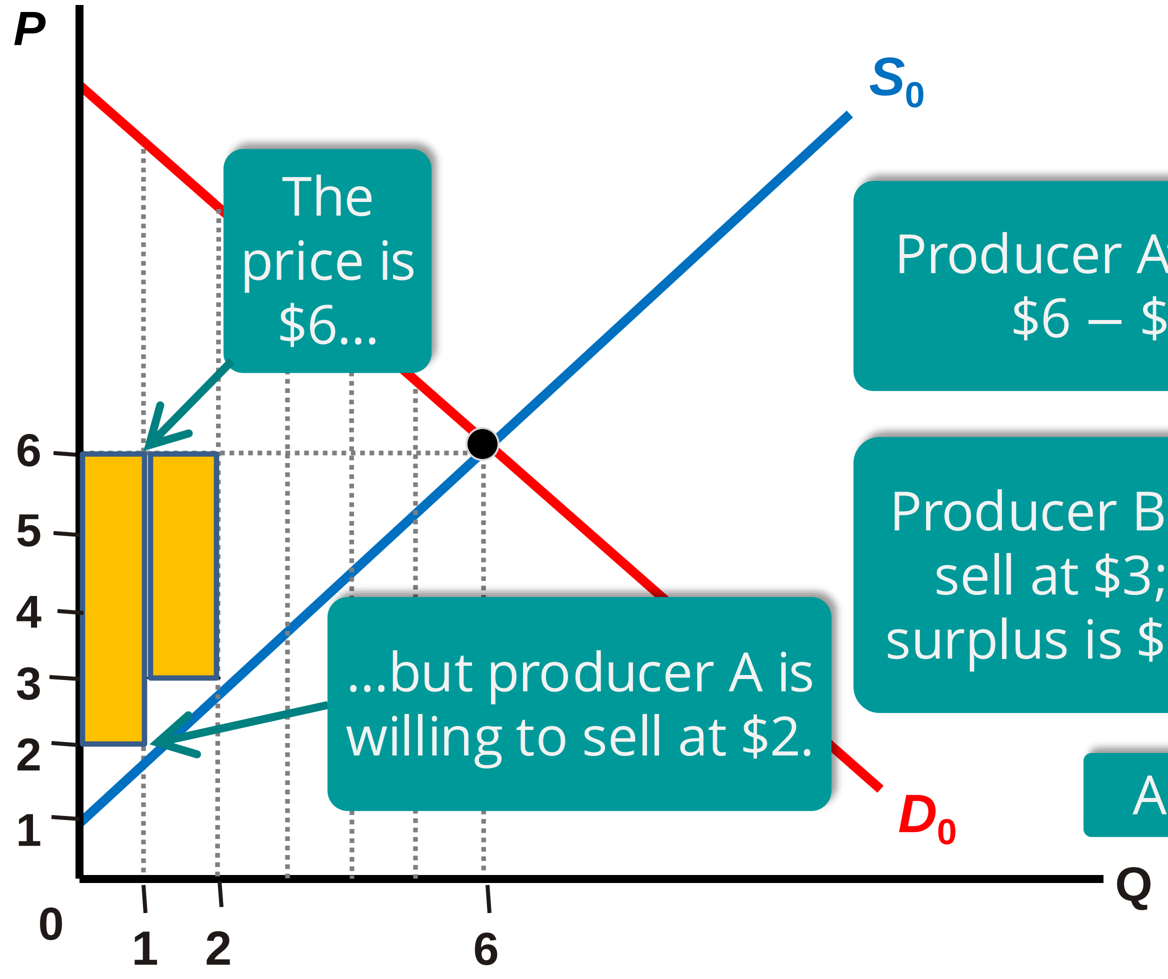
The sky is dumping freezing rain, it’s 11:30 at night, and you’re standing on a street corner 15 miles from home. You open your Uber app and see that the trip is going to cost you roughly $50, as prices are surging 200 percent above normal. But you summon the car anyways. In fact, you might not have hesitated to go as high as $100, just to get dry in that comprising moment.

Laura Bliss, Bloomberg, 15 September 2016

What is your consumer surplus?

What market inefficiency does Uber solve?

INDIVIDUAL PRODUCER



Producer A's surplus is $6 – $2 = $4.

Producer B is willing to sell at $3; producer surplus is $6 – $3 = $3.

And so on…

Four different students each have one textbook they might sell back to the bookstore. Below is each student's reservation price: the lowest price at which they would be willing to sell the book.

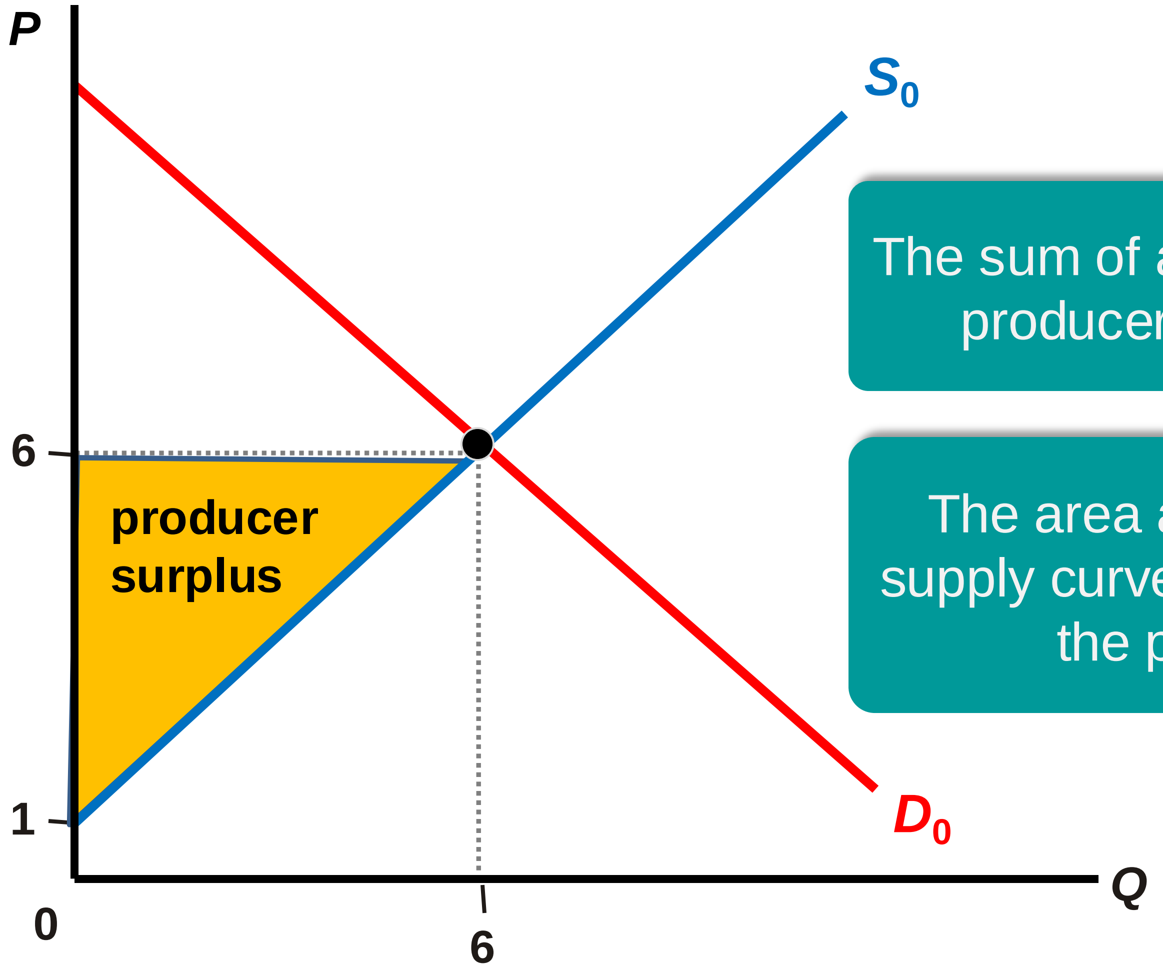
The bookstore is buying textbooks at a price of $25.

Seller	**Seller's Reservation Price**
Alex	$12
Piper	$16
Daya	$22
Red	$30

How many students will sell their textbooks?

a) 0
b) 1
c) 2
d) 3
e) 4

TOTAL PRODUCER SURPLUS

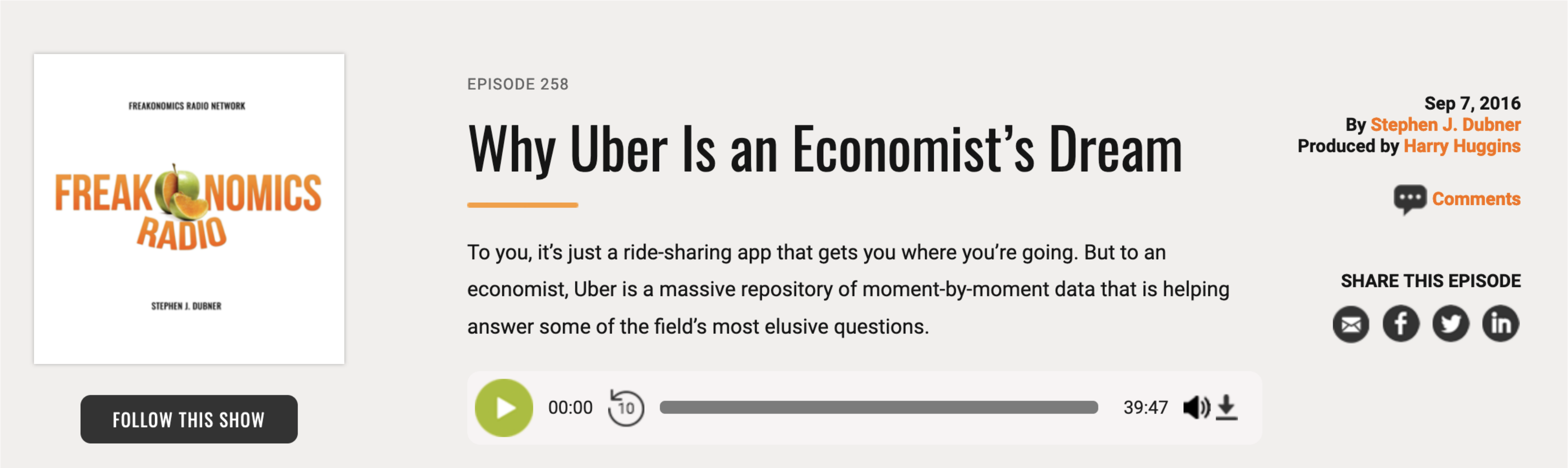


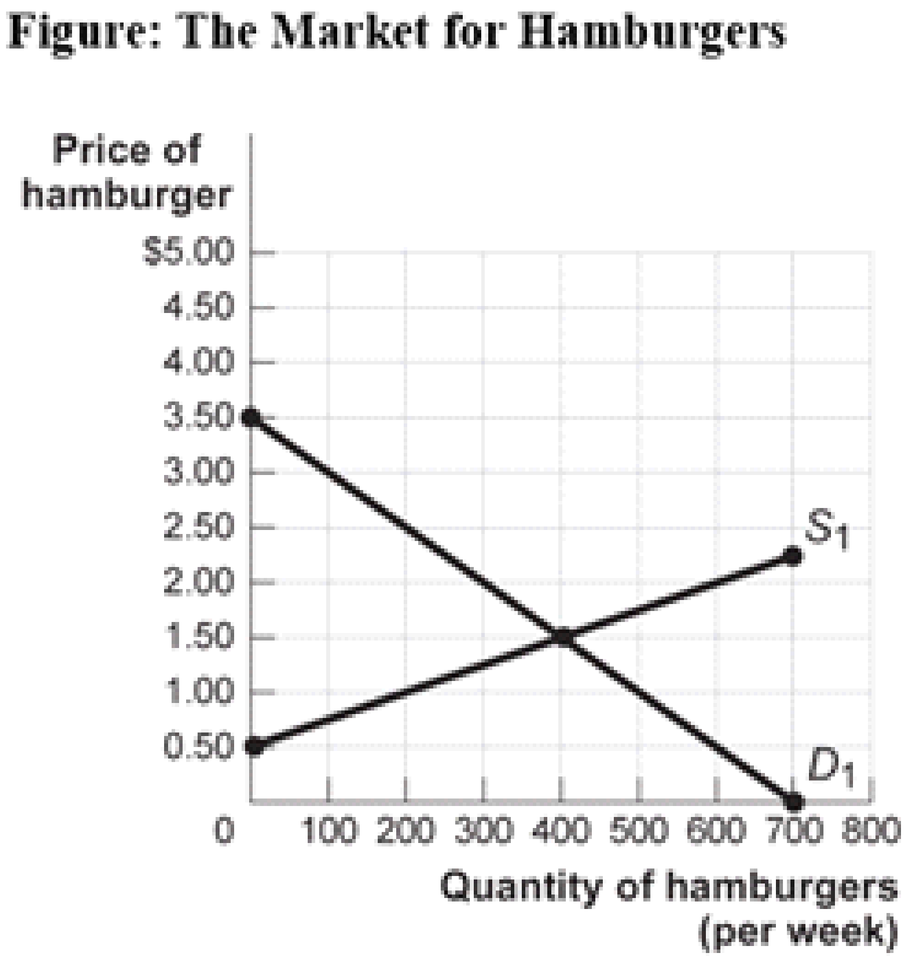
The sum of all individual producer surplus

The area above the supply curve and below the price

How does Uber’s ability to tailor prices affect producer surplus?

Why might Uber be "an economist's dream"?





If 400 hamburgers are sold, producer surplus will equal

a) $650
b) $400
c) $510
d) $200

Figure: The Market for Hamburgers

If the price falls from $1.50 to $1, what is the loss in producer surplus?

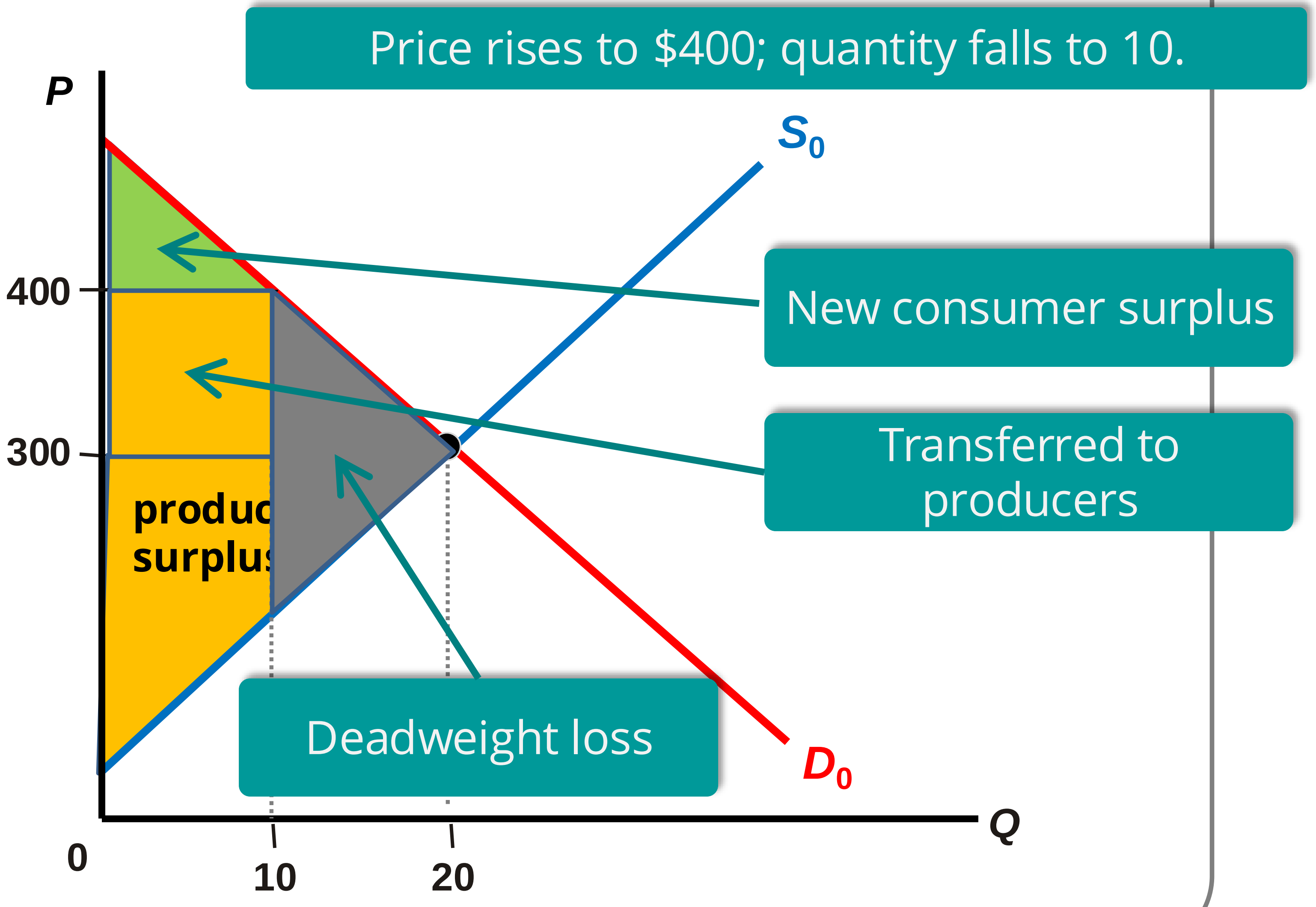
a) $50
b) $45
c) $150
d) $200

TOTAL SURPLUS

When markets are efficient, TS is MAXIMIZED!

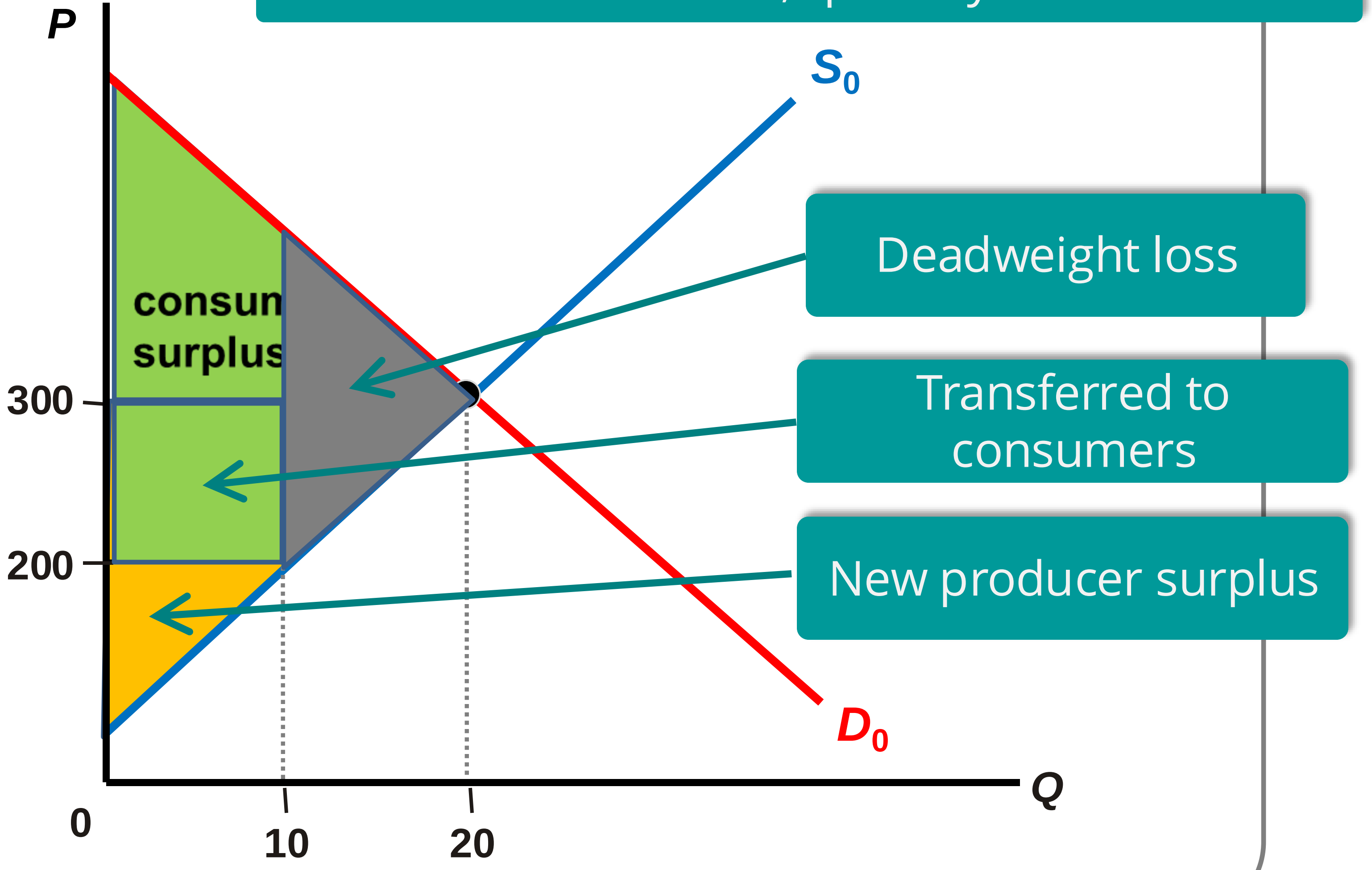
$$\boxed{TS = CS + PS}$$

WHEN PRICES EXCEED



WHEN PRICES ARE BELOW EQUILIBRIUM

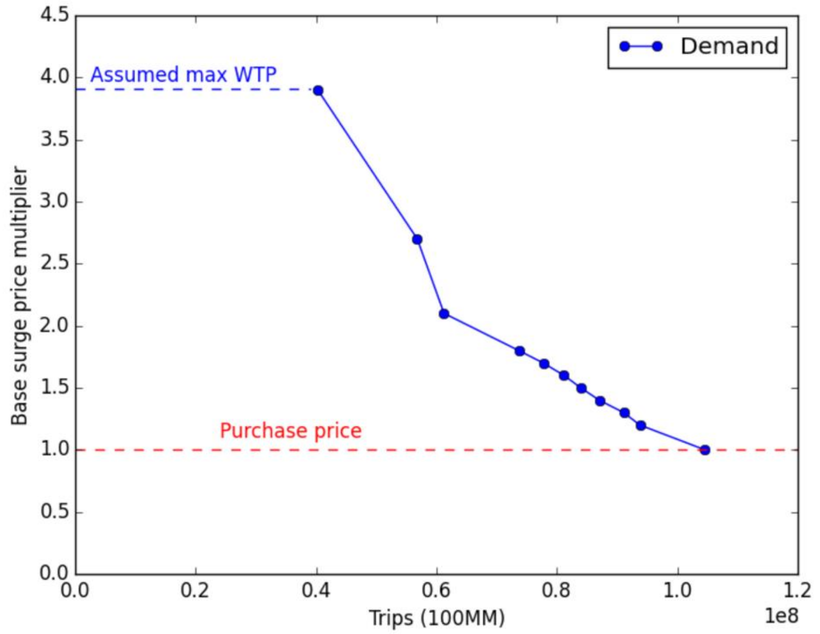
Price falls to $200; quantity falls to 10.



America Would Happily Pay Uber An Extra $7 Billion

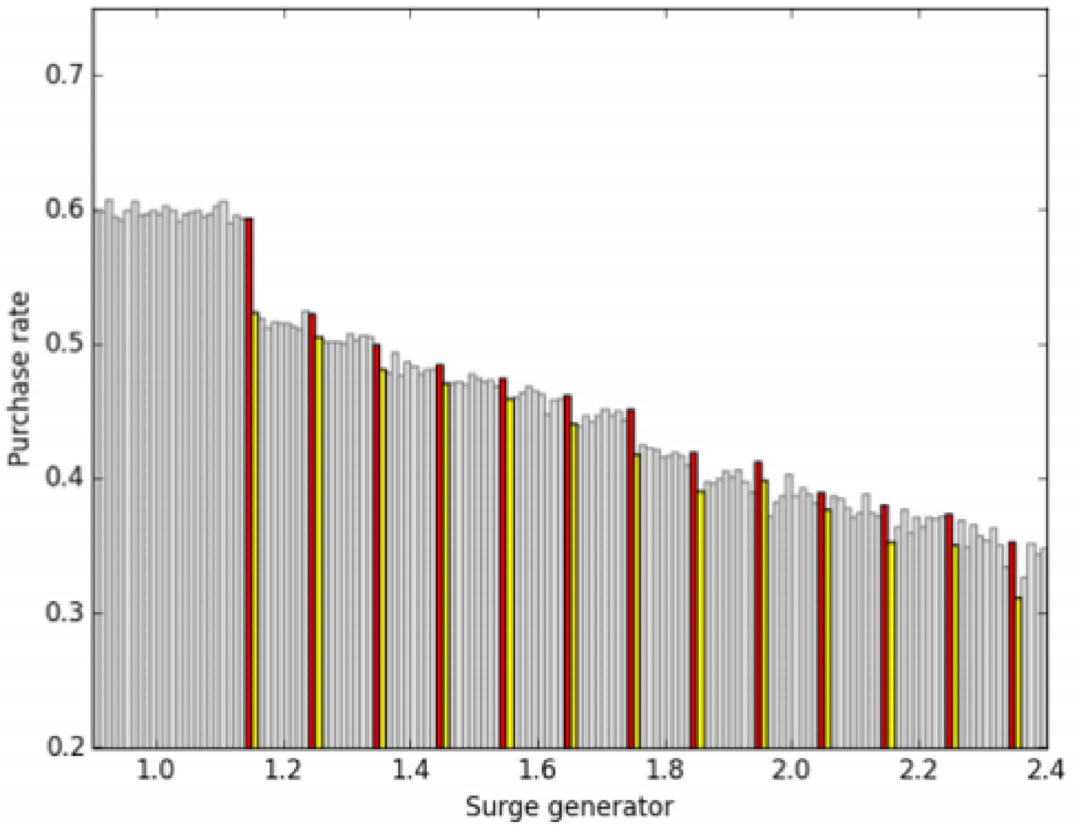
Economists put a (big) number on the ride service's consumer surplus in 2015.

Figure 6: Visual representation of demand curve for transactions at 1.0x



Note: This figure presents a piecewise linear demand curve with jumps at each price discontinuity. The curve is generated from the underlying elasticities estimated for each price discontinuity and for consumers facing transactions at 1.0x.

Figure 5: Request rate drops at pricing discontinuities



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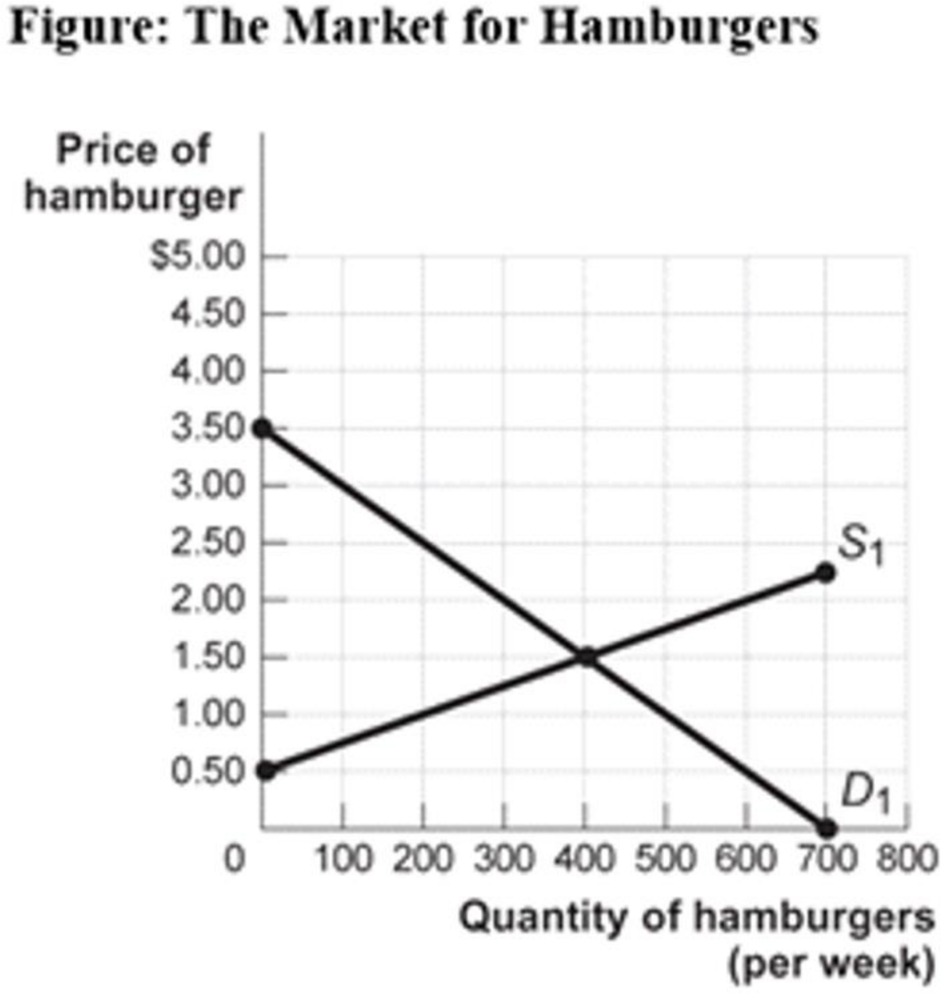

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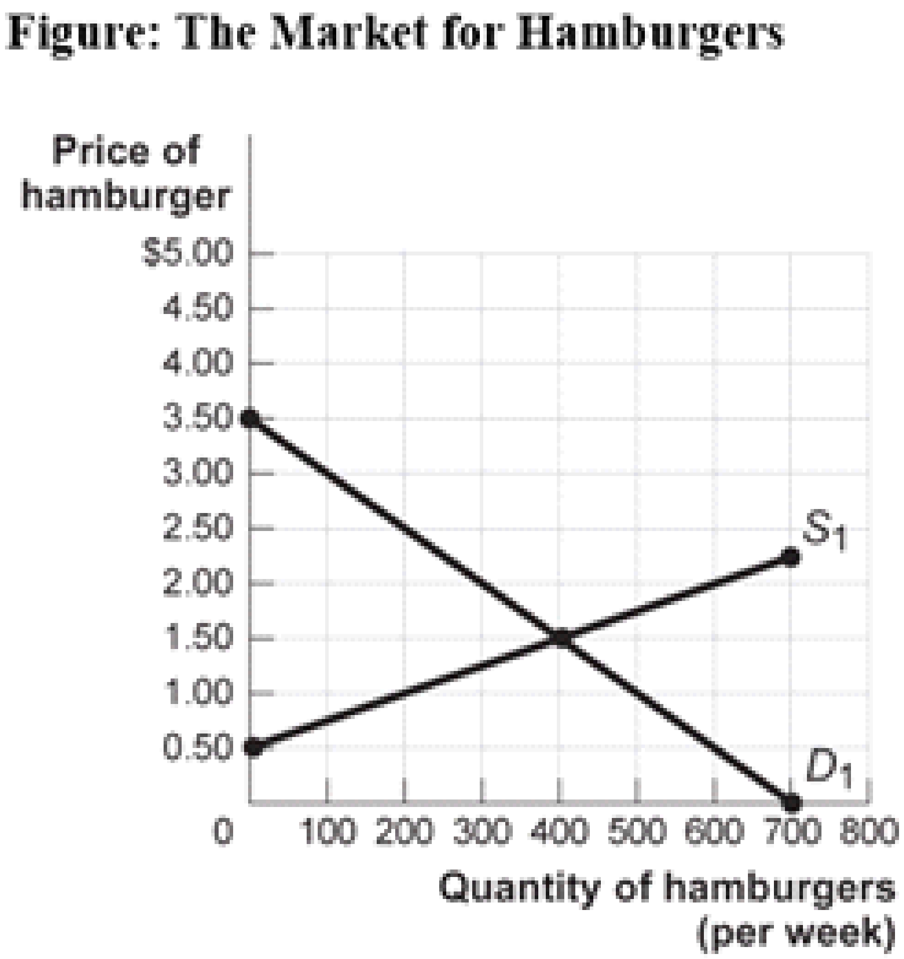

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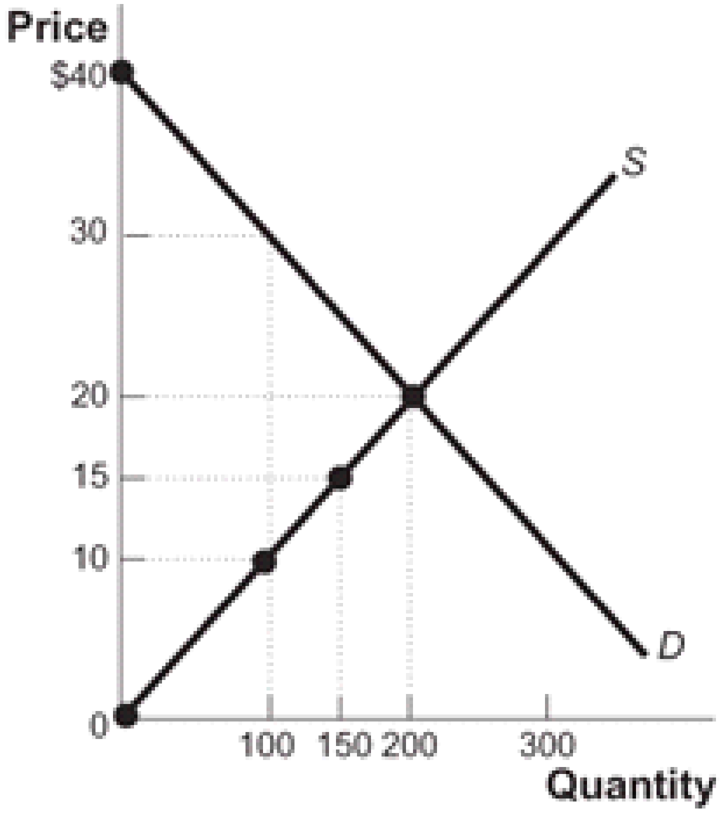
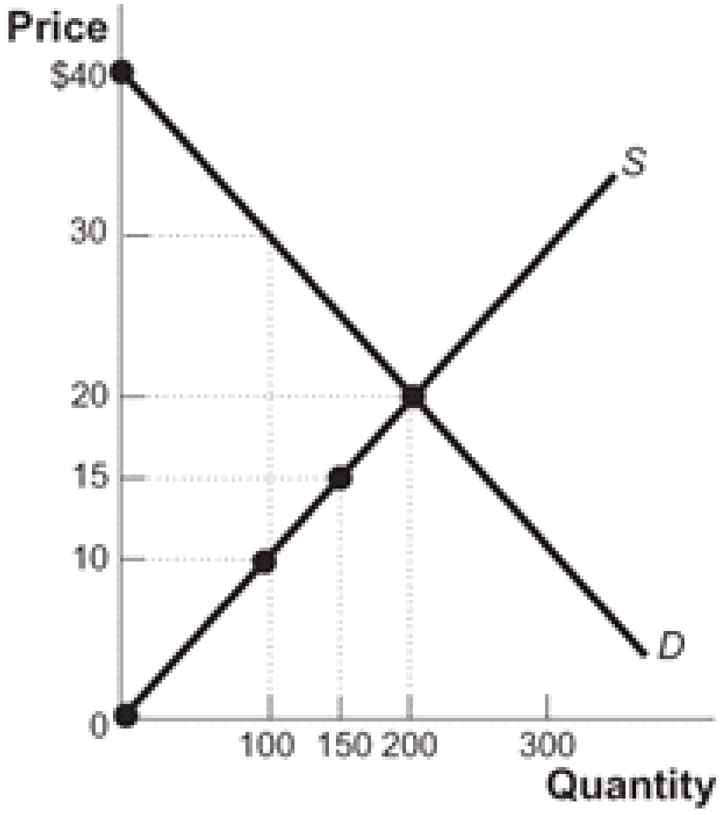


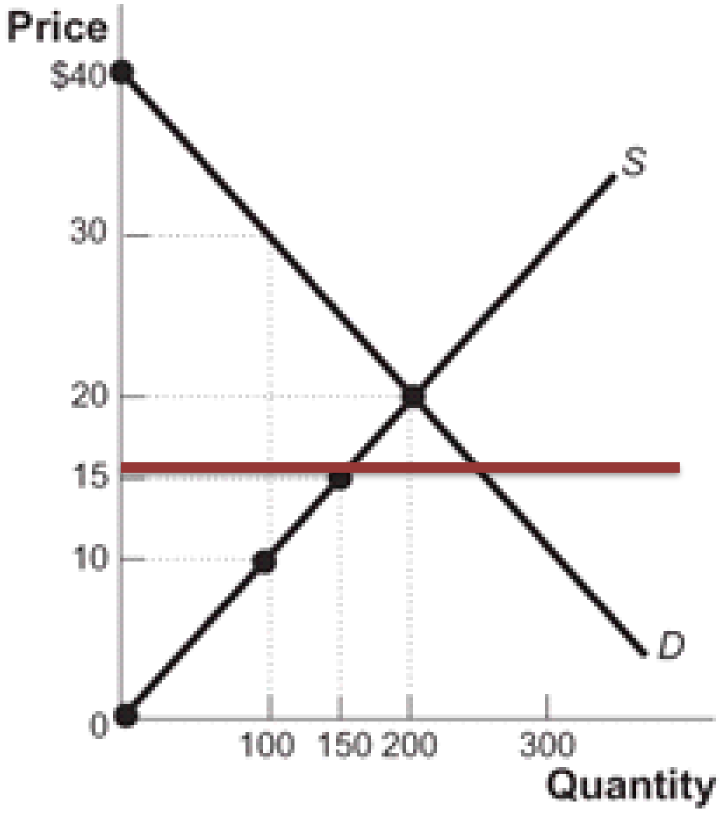
Look at the market for octopus candle holders. What is **producer surplus** at equilibrium?

a) $1000
b) $2000
c) $4000
d) $8000



Look at the market for octopus candle holders. What is **total surplus** at equilibrium?

a) $1000
b) $2000
c) $4000
d) $8000



Amazon caps the price at $15. What is the new **producer surplus?**

a) $1125
b) $1875
c) $2000
d) $2250

Efficiency of Markets

- Competitive markets are usually efficient:
  1. Allocate consumption to the potential **buyers who most value it.**
  2. Allocate sales to potential **sellers** who most value right to sell the good (e.g., **who have the lowest cost**).
  3. They ensure that **all transactions are mutually beneficial**

Why markets *typically* work so well

Well-functioning markets are effective because of:

1. **Property rights:** the rights of owners of valuable items, whether resources goods, to use those items as they chooseor.
2. **Economic signals:** any piece of information that helps people make better economic decisions.

Four sources of market failures

Markets will not be efficient in the presence of **market failures!**

- Lack of competition
- Information not shared by all parties
- Existence of external benefits/costs (externalities)
- Existence of public goods